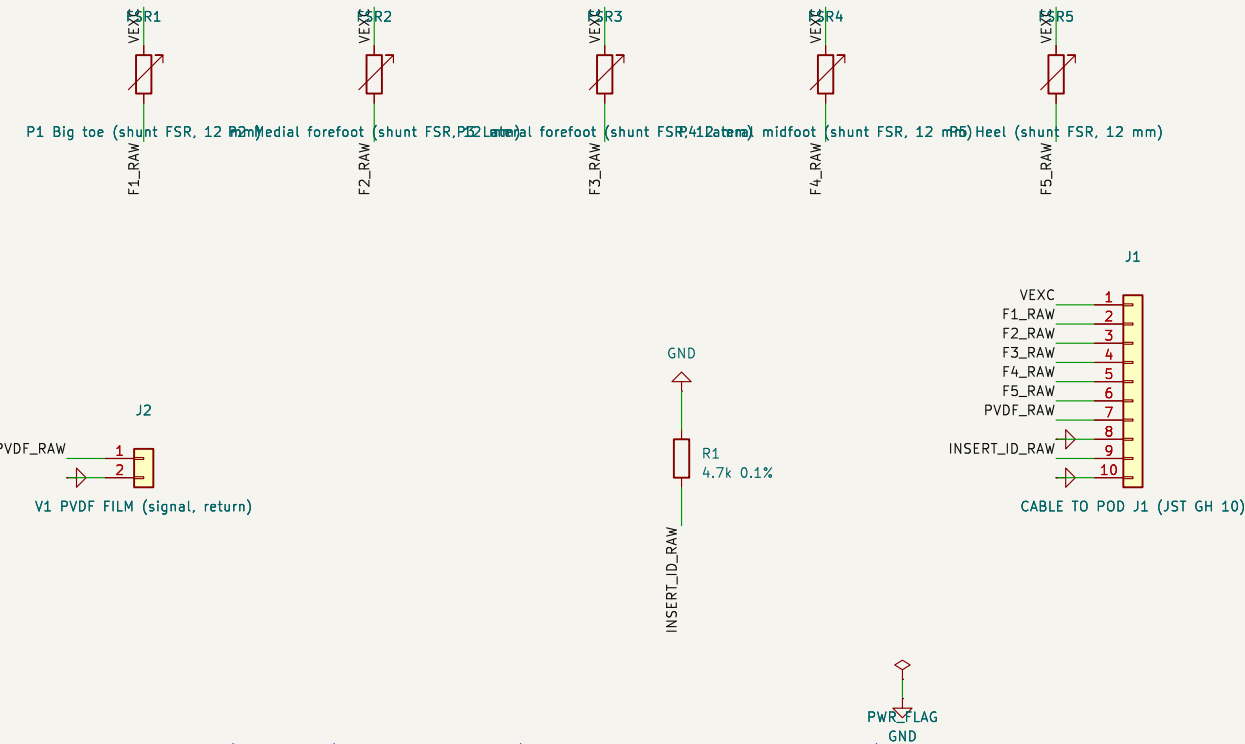


NIVA SENSING INSOLE FLEX – RIGHT MEDIUM – ENGINEERING PROTOTYPE – NOT FOR FABRICATION



Each FSRn: drive comb to VEXC (pod excitation), sense comb to Fn_RAW (pod load resistor + filter to MCP3208, ratiometric).
Five separate drive leads underfoot, joined to VEXC only at the above-collar transition (B.Cu) – no crossings underfoot.
R1 insert ID: $V(ID) = VEXC \times R1 / (10k + R1)$ with pod $R30 = 10k \ 0.1 \ \%$. $4.7k \ 0.1 \ \% = \text{right medium}$.
Right-medium $4.7k \rightarrow 0.320$; left-medium $47k \rightarrow 0.825$ of VEXC. The app must refuse a mismatched side.
FSR response depends on the FSR ink film, spacer and lamination: characterise before any use of the numbers.

Generated by work/build_niva_schematic_revC.py
NOT FOR FABRICATION. Values, protection and layout require bench validation.

NIVA
Sheet: /
File: NIVA-insole-R.kicad_sch

Title: NIVA sensing insole flex | right medium | Rev A (H1)

Size: A4	Date: 2026-09-28	Rev: C – ENGINEERING PROTO
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